

PBA: 1. Course Introduction

Fall 2025

Jaewon (“Jay-one”) Yoo

National Tsing Hua University

Roadmap

1. Welcome and Motivation

Q1: Why learn programming & business analytics?

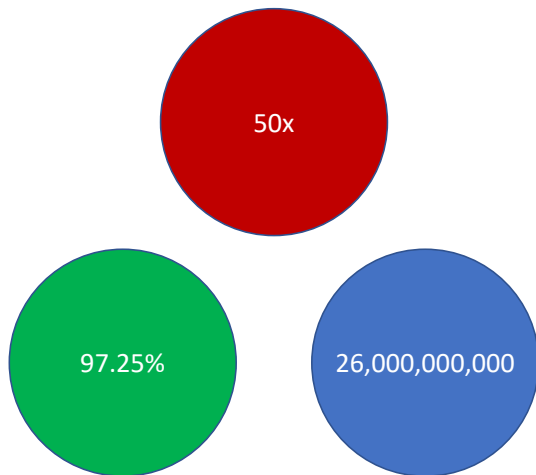
Q2: What problems are data scientists/business analysts working on?

2. Course Overview

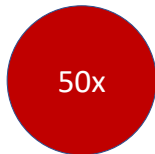
1/ Welcome and Motivation

Why Learn Programming & Business Analytics?

Three Circles w/ Numbers?



The Red Circle: Volume



Total data volume worldwide
in 2020 to increase by 50 folds
compared to only a decade ago!



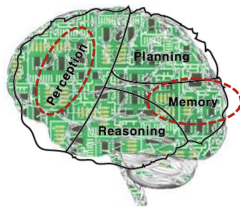
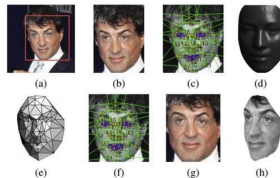
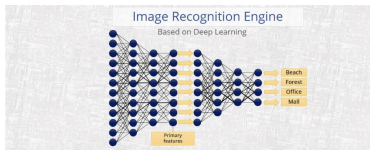
Giga << Tera << Peta << Exa << Zetta << Yotta

- Est. for 2023: 120 ZB (2x compared to 2020; [LINK](#))

The Green Circle: Accuracy

97.25%

Accuracy rate of a face recognition algorithm, Deep Face is 97.25%
(Accuracy of human eyes ~ 97.53%)



- “Facebook’s algo. w/ 97.25% accuracy...” ([Forbes, 2014](#)) $\rightsquigarrow$ **Now?**

Recent Hype on ML/AI

 [MARKETS](#) [BUSINESS](#) [INVESTING](#) [TECH](#) [POLITICS](#) [CNBC TV](#) [INVESTING CLUB](#) [PRO](#)

TECHNOLOGY EXECUTIVE COUNCIL

The ChatGPT AI hype cycle is peaking, but even tech skeptics don't expect a bust

PUBLISHED SAT, FEB 11 2023-11:10 AM EST | UPDATED SAT, FEB 11 2023-2:29 PM EST

 **Eric Rosenbaum**
@ERPROSE

SHARE [f](#) [t](#) [in](#) [✉](#)

Reference: <https://www.cnbc.com/2023/02/11/chatgpt-ai-hype-cycle-is-peaking-but-even-tech-skeptics-doubt-a-bust.html>

Recent Hype on ML/AI

THE WALL STREET JOURNAL.

English Edition | Print Edition | Video | Podcasts | Latest Headlines | More

Subscribe | Sign In

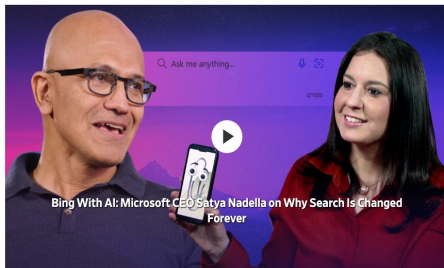
SPECIAL OFFER

Home World U.S. Politics Economy Business Tech Markets Opinion Books & Arts Real Estate Life & Work Style Sports

Search

Microsoft Adds ChatGPT AI Technology to Bing Search Engine

Software company aims to capitalize on OpenAI investment



Microsoft is combining the tech behind OpenAI's ChatGPT with its Bing search engine. In an interview, WSJ's Joanna Stern spoke with Microsoft CEO Satya Nadella about the new tools and how AI is going to change search. (Oh, and Clippy!) Photo illustration: Preston Jessee for The Wall Street Journal

By [Tom Ichniowski](#) Follow

Updated Feb. 10, 2023 7:08 pm ET



Reference: <https://www.wsj.com/articles/microsoft-adds-chatgpt-ai-technology-to-bing-search-engine-11675793525>

Recent Hype on ML/AI

THE WALL STREET JOURNAL.

English Edition | Print Edition | Video | Podcasts | Latest Headlines | More ▾

Subscribe | Sign In

SPECIAL OFFER

Bloomberg

Asia Edition ▾ | Sign In | **Subscribe** | 🔍

• Live Now Markets Economics Industries Technology Politics Wealth Pursuits Opinion **Businessweek** Equality Green CityLab Crypto More ▾

Businessweek
B Schools

ChatGPT Gets an MBA

The AI-powered chatbot did better than expected on a Wharton exam. That's something to get excited about, says the professor behind the experiment.



Photographer: Leon Neal/Getty Images

By **Dimitra Kessenides**
February 4, 2023 at 9:00 PM GMT+8

▶ LIVE ON BLOOMBERG
Watch Live TV >
Listen to Live Radio >

Bloomberg Television

Most Read

Markets
The Devaluation Run in Emerging Markets Is Just Getting Started

Markets
Stocks Are Muted on Inflation Angst; Dollar Gains: Markets Wrap

Pursuits
Rihanna Was Pregnant While Performing Super Bowl Halftime Show

Markets
BlackRock, Pimco Push Back Against Bets Inflation Cooling Fast

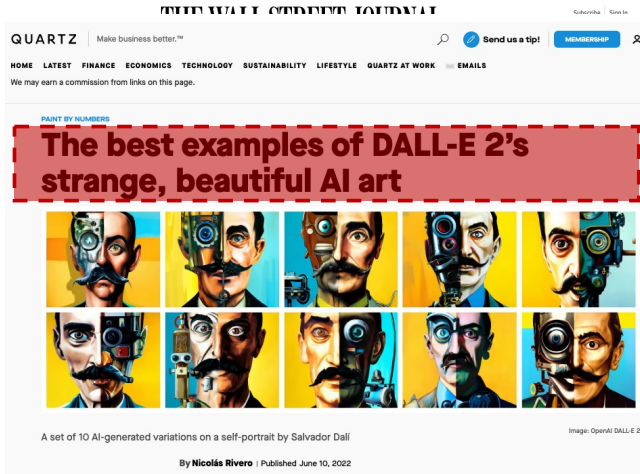
Markets

Reference: <https://www.bloomberg.com/news/articles/2023-02-04/chatgpt-passed-a-wharton-mba-exam-are-professors-worried?leadSource=uverify%20wall>

By [Tom Dotan](#) [Follow](#)
Updated Feb. 10, 2023 7:08 pm ET

Reference: <https://www.wsj.com/articles/microsoft-adds-chatgpt-ai-technology-to-bing-search-engine-11675793525>

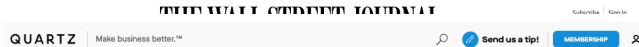
Recent Hype on ML/AI



Reference: <https://qz.com/2176389/the-best-examples-of-dall-e-2s-strange-beautiful-ai-art>

Reference: <https://www.wsj.com/articles/microsoft-adds-chatgpt-ai-technology-to-bing-search-engine-11675793525>

Recent Hype on ML/AI



May 14, 2024

Comparing Text-to-Image Models: DALLE 3 vs Stable Diffusion 3

Explore the performance and differences between DALL·E 3 and Stable Diffusion 3 through a comparison on real-life examples using ChatLabs AI Split Screen.



Reference: <https://writingmate.ai/blog/dalle-3-vs-stable-diffusion-comparison>

Reference: <https://qz.com/2176389/the-best-examples-of-dall-e-2s-strange-beautiful-ai-art>

Reference: <https://www.wsj.com/articles/microsoft-adds-chatgpt-ai-technology-to-bing-search-engine-11675793525>

Recent Hype on ML/AI



Harvard
Business
Review

Sign In

Intellectual Property

Generative AI Has an Intellectual Property Problem

by Gil Appel, Juliana Neelbauer, and David A. Schweidel

April 07, 2023



HBR Staff/Pexels

Reference: <https://hbr.org/2023/04/generative-ai-has-an-intellectual-property-problem>

Recent Hype on ML/AI



Harvard
Business
Review

Sign In

Intellectual Property

Forbes

News Events Lists Life

Sign-in

Subscribe



Uncategorized

Major record labels sue AI music startups for allegedly copying songs to train AI

By Molly Bohannon

June 25, 2024



The three largest record labels are suing two artificial intelligence music companies, alleging the firms committed “mass infringement of



Reference: <https://www.forbes.com/sites/mollybohannon/2024/06/24/major-record-labels-sue-ai-music-startups-for-allegedly-copying-songs-to-train-ai/>



HBR Staff/Pexels

Reference: <https://hbr.org/2023/04/generative-ai-has-an-intellectual-property-problem>

Recent Hype on ML/AI



Harvard
Business
Review

Sign In



THE WALL STREET JOURNAL.

Forbes

SUBSCRIBE

SIGN IN

Subscribe



Uncategorized

Major
copy

By Molly Bohm

TECHNOLOGY | ARTIFICIAL INTELLIGENCE [Follow](#)

OpenAI Says It Can Now Detect Images Spawned by Its Software—Most of the Time

Startup's new tool detects 98% of pictures generated by its DALL-E 3 system, but success drops if the images are altered

By [Deepa Seetharaman](#) [Follow](#)

May 7, 2024 6:00 am ET

Reference
allegedly-c



Share



Resize



Listen (2 min)



DR-



Reference: <https://www.wsj.com/tech/ai/openai-says-it-can-now-detect-images-spawned-by-its-software-most-of-the-time-83011149>

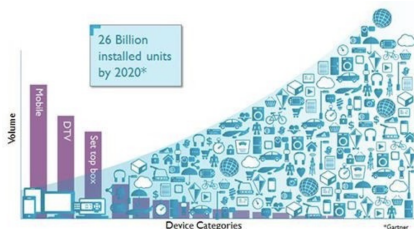
HBR Staff/Pexels

Reference: <https://hbr.org/2023/04/generative-ai-has-an-intellectual-property-problem>

The Blue Circle: Connectivity

26,000,000,000 !!

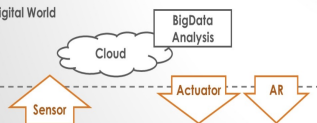
By 2020, ~26 billion devices will be connected to the Internet/network (world population: 7.7 billion)



Future Product = Hardware controlled by Software with Network

Future World = Digital World + Physical World

Digital World



Physical World

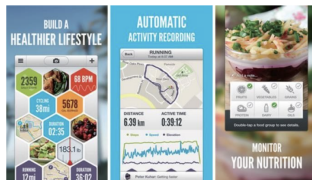
- “In 2023, it’s predicted that there will be more than **43 billion devices** connected to the Internet.” ([Forbes, 2022](#))

Wearable IOT & Sensor Data



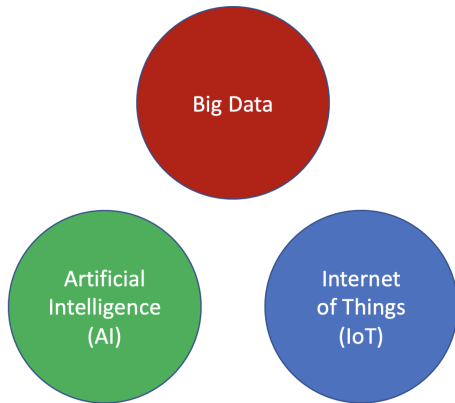
[Reference: Medium, Aug. 2018]

Type	Feature	Usage Purposes
Heart Rate Sensor	Quantify heartbeat per minute	Monitor health status
Pressure Sensor	Measure ambient pressure	Monitor altitude
CdS Sensor	Measure light intensity	Check the presence of light
Infrared Sensor	Detect infrared light emitted by a target	Monitor body temperature
Proximity Sensor	Measure relative distance to an object	Monitor surrounding environment
Gyro Sensor	Measure angular velocity using 3 axes	Monitor intensity of an activity
Velocity Sensor	Measure movement using 3 axes	Monitor intensity of an activity
GPS Sensor	Measure current location	Track location
Temperature Sensor	Measure temperature in the atmosphere	Monitor surrounding environment
Humidity Sensor	Measure humidity in the atmosphere	Monitor surrounding environment
Acoustic Sensor	Measure nearby acoustic waves	Monitor action
Glucose Sensor	Measure glucose level in blood	Monitor health



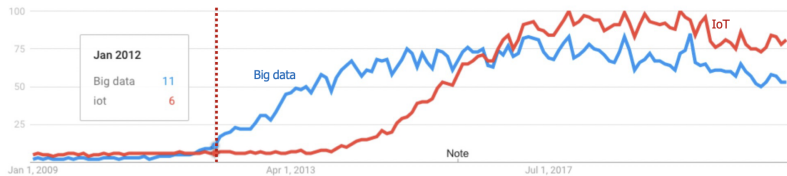
[Reference: Azumio]

Three Most Hotly Debated Techs:



Is Big Data Still an Issue?? ...YES

Google trend's search volume over time:



Why Big Data?

amazon Try Prime Your Amazon.com Today's Deals Gift Cards Sell Help

Shop by Department ▾ Search All recommender system 🔍 Hello, Sign in Your Account

Books Advanced Search New Releases Best Sellers The New York Times® Best Sellers Children's Books Textbooks Textbook Rentals Sell Us Your Books

Look inside ↴



Recommender Systems Handbook Hardcover – October 28, 2010
by Francesco Ricci (Editor), Lior Rokach (Editor), Bracha Shapira (Editor), & 1 more
★★★★★ 1 customer review
ISBN-13: 978-0387858197 ISBN-10: 0387858199 Edition: 2011th

Buy New
Price: \$183.20
29 New from \$166.89 | 16 Used from \$154.00

	Amazon Price	New from	Used from
Kindle 	\$161.99	—	—
Hardcover	\$183.20	\$166.89	\$154.00
Paperback	—	—	—

FREE TWO-DAY SHIPPING FOR COLLEGE STUDENTS
[Learn more](#) **amazon student**

Customers Who Bought This Item Also Bought

Amazon's Recommendation Service



Recommender Systems: An Introduction
Dietmar Jannach
★★★★★ 7
Paperback
\$71.63 



Algorithms of the Intelligent Web
Haralambos Marmanis
★★★★★ 16
Paperback
\$29.13 



An Introduction to Statistical Learning
> Gareth James
★★★★★ 56
#1 Best Seller in Mathematical & Statistical...
Hardcover
\$73.58 

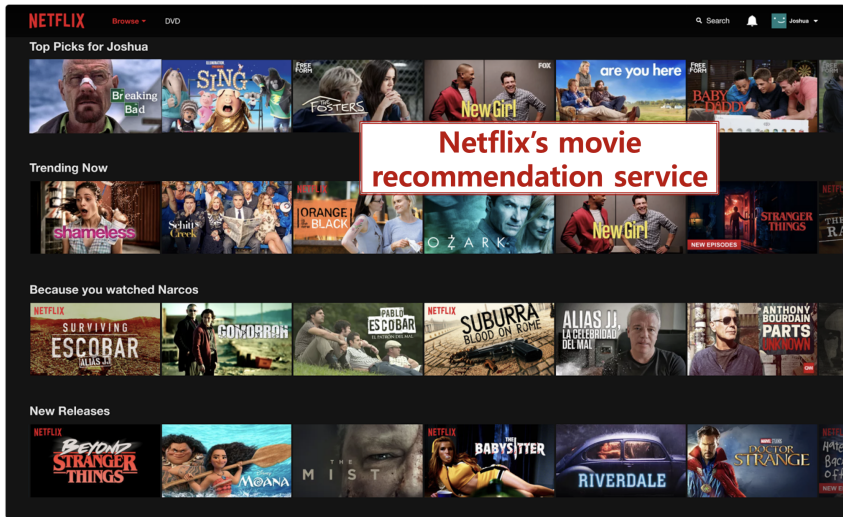


Machine Learning: A Probabilistic...
> Kevin P. Murphy
★★★★★ 42
Hardcover
\$76.50 



Scaling up Machine Learning: Parallel and Distributed Approaches
Ron Bekkerman
Hardcover
\$84.06 

Why Big Data?



Paradigm Shift: From Age of IT to the Age of DT

We are now in the midst of a great infrastructure building project.
We do not know yet the real value of the project.

The name of the project is 'Datafication'.

We will see the world is essentially the universe that is composed of data. If you see the world as the sea of data, it becomes the mental eye to see through every sphere of our life.

Victor Mayer Schonberger (Oxford), BIG DATA: A Revolution That Will Transform How We Live, Work, and Think



Think about it this way:

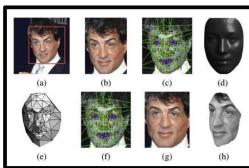
DATAFICATION & ADVANCED ANALYTICS



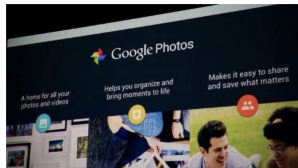
- Facebook has “datafied” our friend network.
- Google has “datafied” our search and information retrieval.
- LinkedIn has “datafied” our professional connections.
- Twitter is “datafying” news and real time information.
- Waze is “datafying” our driving.
- GE is “datafying” all its engines, power plants, and machines.



IBM Watson

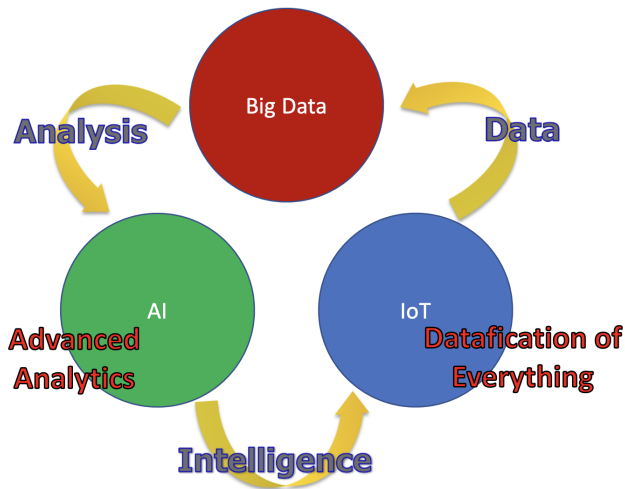


Facebook Deepface



Google Photos

Three Core Tech. in the Era of Digitization



How to Acquire (Big) Data

- “Pick”it up on the Web: **Crawling**
 - Scraping using APIs: Twitter, Facebook
 - Scraping using standard protocols: HTTP, RSS
 - R packages?: rvest, Rcrawler, scraper
- Ask for it: **Institutional Liaison** (OpenAPI)
 - Public Data Portal - Taiwan (<https://data.gov.tw/en>)
 - Public Data Portal - Korea (<https://www.data.go.kr>)
 - Seoul City Open Data Plaza (<http://data.seoul.go.kr>)
- Purchase it: **Data Services**
 - O2 (<http://o2.saltlux.com/o2api>)
 - DataSift (<http://datasift.com/>)

Most Important Skills in the next 10 years?

Google's Chief Economist, Hal R. Varian:



Ability to **Understand** Data



Ability to **Process** Data



Ability to **Extract Value** from Data



Ability to **Visualize** Data



Ability to **Communicate** Data

The Growth and Innovation in “Silicon Alley”



The Growth and Innovation in “Silicon Alley” is spearheaded by FinTech!

Handful of FinTech services (i.e., payment, loan, cloud service) emerged alone in 2015 (100 businesses)

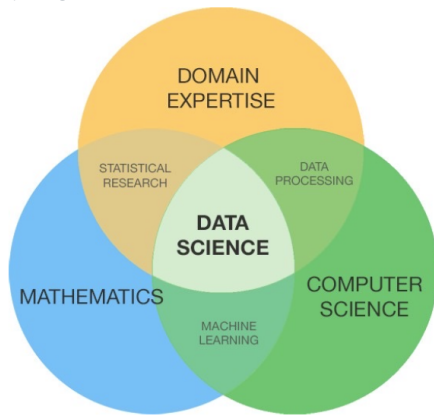
Prof. Peter Blair Henry, Director of NYU Stern School of Business (Weekly BIZ, Feb. 2016)

Q: Which classes are the most popular nowadays?

A: There has been a surge of interest on courses in data science/business analytics. This is because we see a surge of data. Data is also a very important component of FinTech, and we need to understand what we can do with data if we want to be a leader in the industry. In class, students learn how to understand and utilize data,

Data Science? A Convergent Approach to Big Data

Data Science: The study of data collection, storage, processing, and analysis.
cf. Statistics: A branch of applied mathematics that involves gathering, reviewing, and analyzing data.



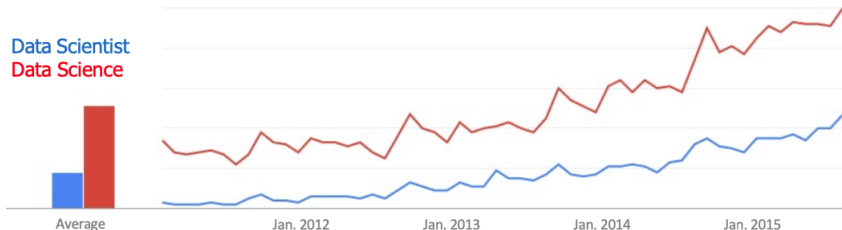
Source: Palmer, Shelly. *Data Science for the C-Suite*.
New York: Digital Living Press, 2015. Print.

Why is Domain Expertise Important?

- How much do you know about the institutional background? Jargons? Trends and paradigms?
- Programs, e.g., R, Python, or Julia, compute the numbers & prints out the results as they are asked.
- $\rightsquigarrow$ **Positing assumptions/conjectures/hypotheses and interpreting the results** are tasks that are really up to us!

Promising Job in the Era of DT: Data Scientist

Data Scientist: The Sexiest Job of the 21st Century (HBR, 2012.10)



Fortune

Supply is down at merely 20% while approximately 4.5 million data scientists are needed globally.

Salaries for entry level data scientists had risen to \$91,000 nationally and \$110,000 in Silicon Valley.

– Glass Door Report

How About Now? A Follow-Up Article in HBR..

Is Data Scientist Still the Sexiest Job of the 21st Century?

by Thomas H. Davenport and DJ Patil

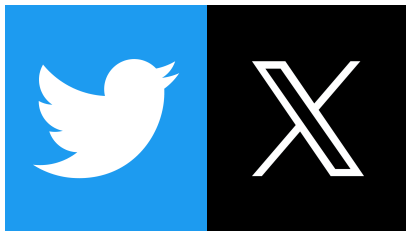
July 15, 2022



HBR Staff/StudioM/Moritz Otto/Getty Images

Reference: <https://hbr.org/2022/07/is-data-scientist-still-the-sexiest-job-of-the-21st-century>

Promising Job in the Era of DT: Data Scientist



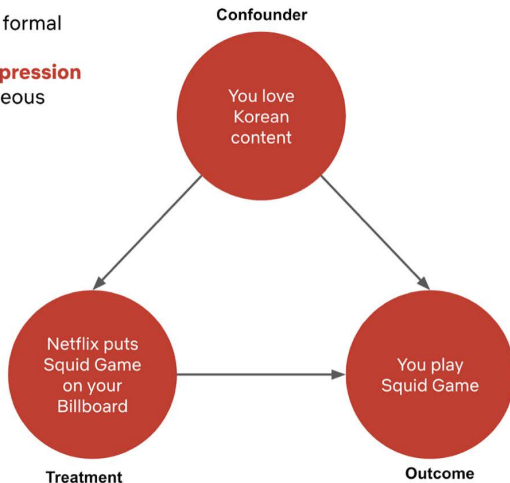
In the Academic ...

Data Scientist (n.): Person who is better at statistics than any software engineer and better at software engineering than any statistician.

What problems are data scientists/business analysts working on?

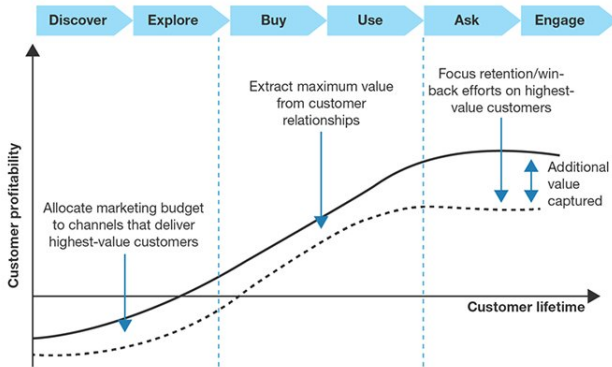
Causality

Causal Inference provides formal tools to tease out the true **incremental** value of an **impression** for each profile: Heterogeneous Treatment Effect (**HTE**)



Compared to machine learning, causal inference allows us to build a robust framework that controls for confounders in order to estimate the true incremental impact to members

Prediction

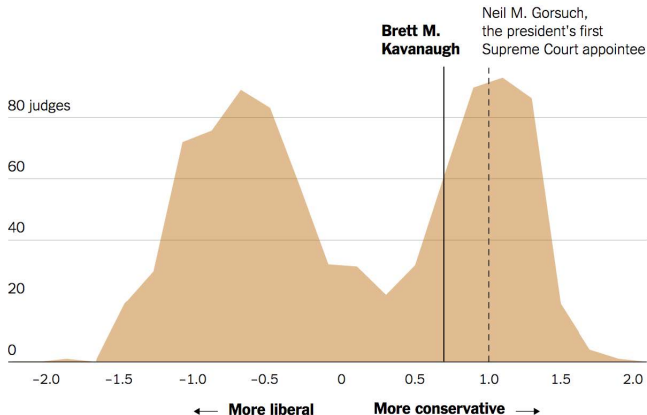


#CXNYC

FORRESTER®

Measurement

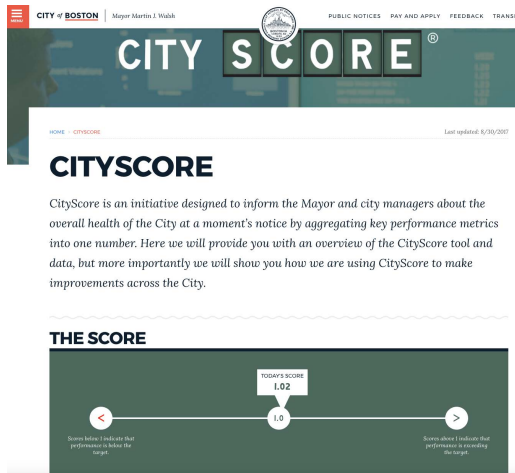
How Kavanaugh's Ideology Compares With Other Federal Judges



Based on the campaign finance scores of all current and former federal district and court of appeals judges nominated since 1980. | Source: [Database on Ideology, Money in Politics, and Elections](#); Adam Bonica, Stanford University Department of Political Science; Maya Sen, Harvard University, Kennedy School of Government; Adam Chilton and Kyle Rozema, University of Chicago Law School.

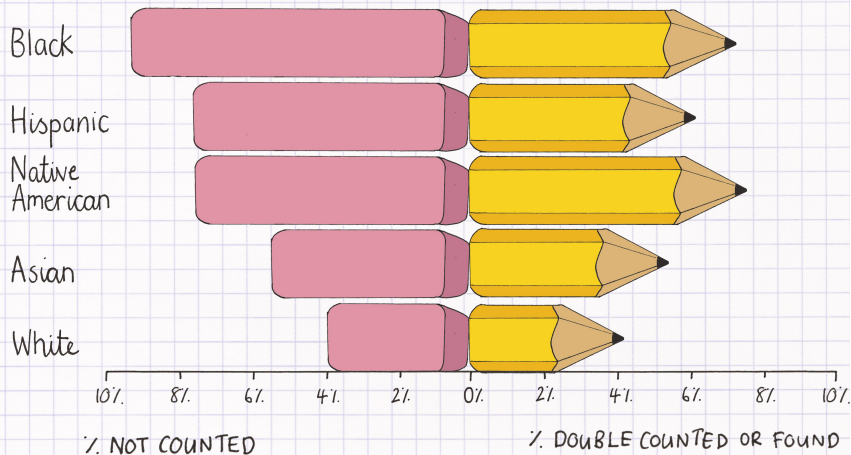
- “You can’t improve what you don’t measure.” - Peter Drucker

Making Organizations Work Better



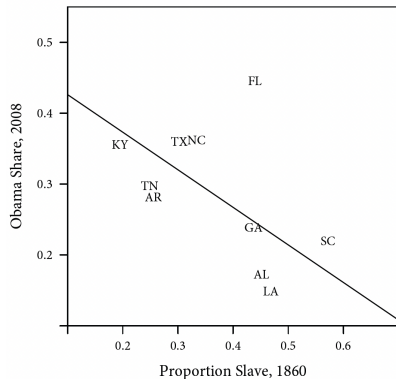
Combining Art and Data to Inform

Who Gets Miscounted In The Census ?

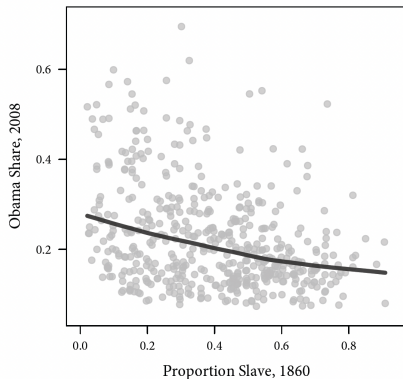


Understanding How the Past Matters

States



Counties



2/ Course Overview

About Me



Figure 1: Illustration of Counterfactuals in CI vs. XAI

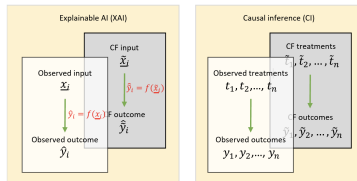


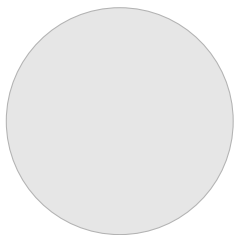
Table 1: Overview of differences in use of counterfactuals

Dimension	Causal Inference	Explainable AI
Purpose	Estimate causal effect	Explain a predicted outcome
Type of Relationship	Causal	Correlation
Quantity of Interest	Difference in outcomes	Input changes
Aggregation Level	Sample	Unit
Modified Object	Model	Data
Assumptions	On data generating function	On given predictive model
Performance Evaluation	Standard error, conf. interval, significance, robustness tests	Feasibility, sparseness, etc.

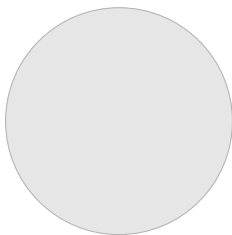
What will you learn in this class?

- Summarize and visualize data
- Wrangle messy data into tidy forms
- Evaluate claims about causality
- Be able to use linear regression to analyze data
- Understand uncertainty in data analysis and how to quantify it
- Use professional tools like R, RStudio, git, and GitHub

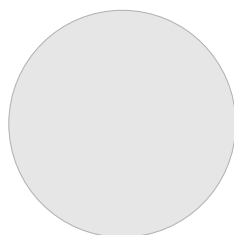
Teaching Assistants & Fellows



John Loui Santiago



Renaldi Ega



TBD

Textbooks

- MD** Ismay, Chester and Albert Y. Kim. 2022. Statistical Inference via Data Science: A ModernDive into R and the Tidyverse.
- QSS** Imai, Kosuke and Nora Webb Willaims. 2022. Quantitative Social Science: An Introduction with Tidyverse, 2022. Princeton University Press.
- IMS** Mine Cetinkaya-Rundel and Johanna Hardin. 2021. Introduction to Modern Statistics. OpenIntro.
- AAG** Jared P. Lander, O'Reilly Media, 2017. R for Everyone: Advanced Analytics and Graphics, 2nd Edition.
- VT** Nathan Yau, John Wiley & Sons, 2011. Visualize This: The Flowing Data Guide to Design, Visualization, and Statistics.

Software Tools

- **R Statistical Programming Language:**

- R is an open-source language that can be used free of charge and can be downloaded from <https://www.r-project.org/>.

- **Integrated Development Environment (IDE):**

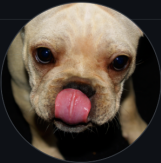
- For improved productivity and convenience, students will also use RStudio, a widely used integrated development environment (IDE) for R which is available at <https://www.rstudio.com/products/rstudio/> for download.
- If you use a MacBook with the M1 chip/Apple silicon, RStudio might not work on your machine. If it does not, you can also use Visual Studio Code (VS Code) to edit your codes which can be downloaded at <https://code.visualstudio.com/>.

git and GitHub



- Other core tools: git and GitHub
 - **Version control system:** an archive of project versions.
 - Allows you to revert back to old versions easily
 - Makes collaboration much more manageable.
- Will feel very odd at first, but you get used to it
- Why learn this now?
 - Knowing git/GitHub is a huge plus for data jobs.
 - Your GitHub profile can showcase your amazing new skills with data!

Sample GitHub profile



Mara Averick
batpigandme

[Follow](#)

810 followers · 137 following

@tidyverse
Missoula, MT
<https://dataandme>
@dataandme@mstdn.social
<https://bsky.app/profile/dataandme.bsky.social>
@dataandme
in/maraaverick

Achievements



Overview Repositories 238 Projects Packages Stars 845

batpigandme / README.md

Hiieee! I'm Mara. 🐾

```
> # I'm the tidyverse developer advocate
> tidyverse::tidyverse_logo()
```



[Followers](#) 810 [Follow @dataandme](#)

I'm also on Mastodon.

Popular repositories

night-owlish Public

🌙 An RStudio, tmThemes, and Ace editor adaptation of @sdras' Night Owl VS Code theme...

🌟 CSS ☆ 160 🗣 59

tverse_contributing Public

📖 "contributing to the tidyverse" from rstudio:conf(2018)

☆ 22

tidymueller Public

thats-not-science Public

Academic Honesty and Plagiarism

- All work submitted for academic evaluation must be the student's own.
 - The penalty for violation of academic integrity will result in a zero for that assignment for the first time.
 - Subsequent violation(s) will result in a failing grade for the course.
- Academic dishonesty comprises of, but is not limited to, the following:
 - **Cheating:** Copying from other students' quizzes and assignments or allowing others to copy from one's own.
 - **Plagiarism:** Using other people's original work without giving appropriate credit or acknowledgment to the authors or sources.
 - **Self-plagiarism:** Submitting a piece of work in more than one course without the explicit permission of the instructors involved.
 - **Misrepresentation of authorship:** Submitting work as one's own, which has been prepared by or purchased from others. Students will be asked to upload their submission materials to Canvas.

AI Use Policy**

- To maintain academic integrity, students who choose to use AI tools (e.g., GPT-4o, Claude 3.5, Gemini) must adhere to the following guidelines:
 - **Thorough Explanation:** Students must provide a detailed explanation of how the AI tool was used in the completion of the assignment.
 - **Original AI Responses:** Alongside their submission, students are required to submit the original responses generated by the AI tool.
 - **Critical Evaluation and Personal Input:** Students should critically evaluate the AI-generated content and clearly indicate the portions of the work that have been modified or expanded upon with their own insights and understanding.
 - **Proper Attribution:** Proper attribution should be given to the AI-generated content used, indicating clearly the sections that are AI-generated.
- Remember, attempting to cheat the system by **heavily relying on AI-generated content without substantial personal input** is ultimately the student's loss!

Final Project (25%)

- Open-ended data analysis project on a topic of your choice (e.g., voting trends, racial disparities in policing, NBA salaries, box-office economics of Marvel movies).
- Steps:
 1. Develop a clear research question
 2. Collect and prepare relevant data
 3. Apply methods from the course
 4. Present results in a clear and accessible way
- Goal: produce a polished, portfolio-ready project demonstrating your data skills.
- Deliverables include: research question, dataset overview, visualizations, regression analysis with plain-language interpretation, discussion of limitations, and a GitHub repo with all code.
- Milestones: Repository (Week 4), Data and Proposal (Week 6), First Visualization (Week 9), First Analysis (Week 11), Final Report (Week 15).

Assignments (20%)

- Five assignments spaced throughout the semester.
- Assignments connect directly to class topics and typically involve running and reporting a statistical method.
- Example: after covering OLS, you will analyze a provided dataset and report the results in a short article-style write-up.
- Deliverables: 2–3 page report generated using R Markdown/Quarto.
- Each assignment = 4% of course grade. Late penalty: –1 percentage point per day (e.g., 2 days late caps at 2% out of 4%).

Problem Sets (20%) & Quizzes (15%)

- **Problem Sets:**

- Regular problem sets drawn from recommended textbooks.
- Designed to reinforce understanding of key statistical/analytical concepts.

- **Quizzes & In-class Exercises:**

- Short quizzes on course material to maintain learning momentum.
- In-class group exercises to practice methods hands-on.

Participation (10%) & Attendance (10%)

- **Participation:**

- Stay engaged and active in discussions.
- Positive and thoughtful contributions improve your score.
- Be ready to share your perspectives and ideas.

- **Attendance:**

- Attend every class with laptops ready for exercises.
- Notify me and the TAs in advance if you must miss a class.
- You remain responsible for all missed material and required work.

Grading Policy

- **Breakdown:**
 - Final Project: 25%
 - Assignments: 20%
 - Problem Sets: 20%
 - Quizzes & Exercises: 15%
 - Participation: 10%
 - Attendance: 10%
- **Bump-up:** Students close to cutoffs who stay engaged and make consistent contributions may receive a grade boost.

Have a Great Weekend! :)

Contact Information:

jaewon.yoo@iss.nthu.edu.tw

<https://j1yoo.github.io/>

